

Review

Transferability and Robustness in Proximal and UAV Crop Imaging

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Abstract

AI-driven imaging is becoming central to crop monitoring, with proximal and unmanned aerial vehicle (UAV) platforms now routinely used for disease and stress detection, yield estimation, canopy structure, and fruit counting. Yet, as these models move from plots to farms, the main bottleneck is no longer raw accuracy but robustness under distribution shift. Systems trained in one field, season, cultivar, or sensor often fail when the scene, sensor, protocol, or timing changes in realistic ways. This review synthesizes recent advances on robustness and transferability in proximal and UAV imaging, drawing on a corpus of 42 core studies across field crops, orchards, greenhouse environments, and multi-platform phenotyping. Shift types are organized into four axes, namely scene, sensor, protocol, and time. The article also maps the empirical evidence on when RGB imaging alone is sufficient and when multispectral, hyperspectral, or thermal modalities can potentially improve robustness. This serves as a basis to synthesize acquisition and evaluation practices that often matter more than architectural tweaks, which include phenology-aware flight planning, radiometric standardization, metadata logging, and leave-one-field/season-out splits. Adaptation options are consolidated into a practical symptom/remedy roadmap, ranging from lightweight normalization and small target-set fine-tuning to feature alignment, unsupervised domain adaptation, style translation, and test-time updates. Finally, a benchmark and dataset agenda are outlined with emphasis on object-oriented splits, cross-sensor and cross-scale collections, and longitudinal datasets where the same fields are followed across seasons under different management regimes. The goal is to outline practices and evaluation protocols that support progress toward deployable and auditable systems, noting that such claims require standardized out-of-distribution testing and transparent reporting as emphasized in the benchmark specification and experiment suite proposed here.

Keywords: machine learning; deep learning; digital agriculture; datasets; domain shift



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1. Introduction

Artificial intelligence (AI) and computer vision (CV) are increasingly expected to guide decisions over the agricultural season, from seedling emergence and weed control to nutrient diagnostics, yield forecasting, canopy health, and fruit detection. Fulfilling this promise requires large, high-quality datasets that reflect real operational variability (seasons, sites, cultivars, management, and sensors). This is hard to achieve in agriculture, as uncontrolled and unstructured conditions introduce substantial variation into any measurement. In practice, data distributions can shift sharply across locations, sensors, dates, management regimes, and growth stages. As a result, the central obstacle to reliable

deployment is achieving robustness under distribution shift [1–3]. In practice, models trained in one field, season, cultivar, sensor, or management regime often fail under new illumination, phenology, background structure, or scale [4–8]. Recent studies have begun to quantify these gaps and demonstrate practical solutions across crops, traits, and imaging setups [9–12]. Accordingly, the practical question is not whether proximal/UAV models can reach high accuracy in-domain but how reliably they carry that performance across the routine changes that occur between plots, farms, sensors, and seasons.

This article focuses on in-field data acquisition with proximal and unmanned aerial vehicle (UAV) imaging, including red–green–blue (RGB), multispectral (MS), hyperspectral (HS), and thermal modalities. In practice, robustness is often won or lost upstream, before any model is trained. These platforms directly expose models to the operational variability that most often breaks transferability (site, cultivar, management, illumination, and phenology) and therefore provide the most informative setting for studying robustness under distribution shift. Throughout, satellite imagery is not treated as a primary sensing modality but rather is discussed only when it is explicitly coupled to proximal/UAV data in cross-scale transfer scenarios (e.g., UAV-to-satellite mapping) or when it provides methodological context.

In this review, we use the following terms consistently: *generalization* refers to performance on new samples drawn from the same data-generating process as training (in-distribution); *robustness under distribution shift* refers to maintaining acceptable performance when the test distribution differs from training due to realistic changes in scene, sensor, protocol, or time (out-of-distribution robustness) and is the primary deployment objective here; *transfer learning* (TL) denotes reusing knowledge from a source dataset or task to improve a target crop-imaging task, typically with at least some target labels; and *domain adaptation* (DA) is a TL setting where the task is unchanged but the input domain shifts, with supervised or unsupervised (UDA) variants depending on target-label availability. Throughout, we use *transferability* to mean the empirical degree to which a trained model maintains performance across domains with zero or minimal updating.

On the model side, transfer learning (TL) and physics-guided pretraining form a second pillar of robustness. A practical strategy is to start from a broadly trained reference, such as an ImageNet-style or self-supervised RGB model that captures generic texture and structure cues, and then to fine-tune on a smaller dataset of field images to adapt to local crops and conditions [10,13,14]. When spectral or physiological traits are central, using physics to simulate how crops would reflect light under different leaf area index (LAI), chlorophyll, water content, soil backgrounds, and sun/sensor angles, thus generating better pretraining signals, can further prepare the network before field fine-tuning [15–18]. To bridge site and sensor gaps in practice, input data can be stabilized with color constancy and exposure/white-balance control for RGB or via band selection and reweighting that emphasize invariant spectral cues for MS and HS data [9,19–21]. Representations can also be aligned so that features from different cameras and systems become comparable, for example, using spectral standardization paired with latent-space or transfer component analysis (TCA [22]) alignment for hyperspectral data [23,24]. Finally, fusing complementary geometry or thermal cues helps regularize predictions under changing illumination or canopy structure, improving transfer to new fields and seasons for water-related and stress traits [25]. These ideas reappear in Section 6 as part of the adaptation roadmap from low-cost updates to more involved alignment and updating strategies.

In deployment, conditions often shift faster than labels can be collected, so a practical sequence is to start with a small, spatially diverse target set and perform fine-tuning with roughly 10 to 30% of target samples. This is often enough to adapt the model to the new region, cultivar, or management regime without retraining from scratch [4,10,20].

Such simplified fine-tuning has been effective for head and bunch detection, stress and yield mapping, and nitrogen-related trait detection across cereals, orchards, and soybean breeding trials [7,10,14]. When labels are scarce or unavailable, domain adaptation (DA) and unsupervised domain adaptation (UDA) provide additional options. First, shallow feature alignment methods (e.g., TCA or spectral-response-aware band transforms) align feature distributions and band spaces across sites and sensors, stabilizing classifiers with minimal supervision [23,24,26]. Next, teacher–student consistency schemes and related UDA strategies can be applied so a source-trained teacher guides a student on unlabeled target imagery, a pattern explored for multisensor nitrogen and disease mapping [27,28]. Finally, appearance differences can be addressed directly with pixel-level style translation that makes source images “look like” the target domain (e.g., CycleGAN [29] and CUT-style transforms [30,31]) while preserving geometric content for detection and segmentation [32–34]. These techniques can be paired with classic cross-sensor band correction and multisensor fusion, which reduce input drift and emphasize the most informative traits [20,26].

Crucially, none of these remedies is meaningful unless evaluation protocols actually match the intended deployment gap. Evaluation design dictates whether reported accuracy reflects deployable performance. Studies that compare random splits with spatial or leave-one-field-out validation show that naive pixel- or plant-level splits artificially improve accuracy by leaking near-identical backgrounds and illumination into both training and test sets, thus overestimating transferability [6,7]. In practice, spatially aware protocols tend to yield more conservative and realistic results, especially when test parcels differ in soil, management, or phenology from the training set [10,35]. Open benchmarks make these gaps visible at scale. For example, the Global Wheat Head Detection (GWHD) dataset frames cross-site wheat head detection with standardized train/test farms for out-of-domain generalization [8], and MinneApple emphasizes fruit detection under heterogeneous orchards, cultivars, and lighting to probe robustness beyond random splits [7]. Targeted domain-shift evaluation sets go further by isolating specific factors. Grape-bunch datasets with controlled temporal, viewpoint, and variety changes reveal how performance decays with camera pose or cultivar shifts and which adaptations recover it [14]. Weed, disease, and weather-driven benchmarks quantify how illumination, management, and season interact in segmentation and grading tasks [12,27]. Cross-site root and soil imaging benchmarks separate lab/field and location effects to test whether models truly travel across acquisition setups [36,37]. Together, these designs help identify failure modes early and guide the choice of adaptation tactics that work for real deployments [6,35].

The remainder of the paper proceeds from defining shift types and how to test them, to summarizing empirical evidence by modality, and then to providing acquisition/evaluation guidance and an adaptation roadmap that links common failure modes to practical remedies. This review synthesizes robust evidence for proximal and UAV imaging across agriculture, including field crops, orchards, and protected environments. Considering the four axes of scene, sensor, protocol, and time, the contributions are as follows:

- Practical definition of robustness under distribution shift for agricultural imaging and a field-focused taxonomy of shift types (scene, sensor, protocol, and time) tailored to proximal and UAV data (Section 3);
- Mapping of empirical evidence across RGB, MS, HS, and thermal sensors for field crops, orchards, and protected environments, highlighting when additional modalities consistently improve robustness (Section 4);
- Acquisition and standardization checklist covering flight timing, exposure and white balance, ground-sampling consistency, radiometric handling, and spatially aware

sampling, plus an evaluation protocol template centered on leave-one-field-out and independent-site tests (Section 5);

- Consolidation of adaptation options (lightweight normalization and augmentation, small target-set fine-tuning, feature alignment, UDA, style translation, test-time updates, and uncertainty calibration) into a practical decision support that links common shift effects to low-cost solutions (Section 6);
- Short guide to the most useful public and emerging benchmarks (e.g., GWHD for wheat heads, MinneApple for fruit, etc.), pinpointing where empirical evidence on cross-domain robustness is thin and outlining a focused roadmap for future datasets and experiments (Section 7).

Taken together, these contributions aim to move robustness under distribution shift from a loosely stated aspiration to a set of concrete acquisition, evaluation, and adaptation practices that can be adopted in real-world agricultural imaging pipelines.

This article focuses on robustness under domain and distribution shift for proximal and UAV crop imaging pipelines. We do not attempt exhaustive coverage of purely satellite-based remote sensing, non-imaging sensor streams (e.g., point spectroscopy/IoT without a CV component), or non-crop domains such as livestock and forestry. Our emphasis is on practices and methods supported by field-relevant evidence under shift-aware evaluation.

In this review, UAV imaging refers to aerial acquisition from drone platforms (typically nadir or oblique mosaics), while proximal imaging refers to ground-based close-range sensing (handheld, rig, and phenotyping platforms). When recommendations are platform-specific, they are labeled explicitly as UAV- or proximal-oriented.

2. Literature Search and Article Selection

To ensure transparency and reproducibility, this review followed a structured procedure to identify, screen, and characterize the literature on robustness under domain and distribution shift in proximal and UAV imaging for crop monitoring. The methodology comprised three main steps: a systematic search across major academic databases, screening and application of explicit inclusion and exclusion criteria, and structured characterization of the final corpus of studies. The searches were executed on 8 October 2025 and covered records indexed from database inception through the search date, with no publication-year filter applied.

The search strategy was designed to identify work at the intersection of domain or distribution shift, transfer learning, and related robustness concepts, and proximal or UAV imaging for major field crops and orchards using RGB and multisensor data. The database query was constructed in two layers. First, the “title” field was required to contain at least one term directly related to robustness and transfer, namely “domain shift”, “distribution shift”, “domain adaptation”, “transfer learning”, “model transfer”, “calibration transfer”, “transferability”, “OOD”, or “out of distribution”. Second, the “title–abstract–keywords” fields were required to jointly mention at least one target crop (wheat, rice, maize/corn, soybean, apple, citrus, mango, tomato, grape, or pear), at least one imaging or vision-related term (image, computer vision, remote sensing, hyperspectral, multispectral, RGB, RGB-D, LiDAR, or thermal), and at least one proximal or near-field platform term (UAV/UAS/drone, gantry or rail system, proximal or handheld imaging, ground robot/UGV, tractor or sprayer, or greenhouse/in-field phenotyping). Only records satisfying both layers were retrieved.

This query restricts the “title” field to core robustness terms (e.g., domain adaptation, transfer learning, calibration transfer, transferability, and out of distribution) so that domain or distribution shift is central to the article, while the “title–abstract–keywords” fields constrain crop types, sensing modalities, and platforms. An equivalent query, with the

same logical structure, was implemented in Web of Science (WoS). The Scopus query returned 98 records. The WoS query returned 62 records, of which 54 overlapped with the Scopus results. After aggregating and deduplicating the results from both databases, an initial pool of 106 unique articles was obtained.

The 106 records were then screened using explicit eligibility criteria based on title, abstract, and, when necessary, full text. Inclusion criteria were as follows: (i) written in English; (ii) focused explicitly on domain or distribution shift (or closely related transfer/robustness concepts) in the context of crop imaging; and (iii) used proximal or UAV imaging for the targeted crop types and sensing modalities defined by the query. Works were excluded when robustness or transferability was only mentioned tangentially, when the imaging component was not proximal or UAV (such as purely satellite-based studies or non-imaging sensors without a CV component), or when the main emphasis was unrelated to distribution shift (e.g., generic yield mapping without cross-site or cross-sensor evaluation). After applying these criteria, 42 articles remained and were retained as the core corpus for the review. All synthesis in Sections 3–7 is grounded primarily in these 42 studies.

To reduce the risk of missing relevant work, the reference lists of all 42 core articles were examined (backward snowballing). Each cited paper was checked against the same inclusion criteria. Because the database query constrains crops and imaging modalities, the resulting core set is treated as a focused corpus (major field crops and orchards under proximal/UAV imaging) rather than an exhaustive census of all agricultural sensing studies. No additional article satisfied these criteria beyond the initial 42. It is worth pointing out that, throughout the thematic sections, additional references are cited when they provide relevant context or methodological detail (for example, agronomic background, related modeling strategies, or sensor-specific discussions). These context papers are not counted as part of the core set and are not used in quantitative tallies of robustness evidence, but they help situate the core findings within the broader agricultural AI and phenotyping literature.

Each of the 42 core articles was coded using a structured template capturing the following: crop or domain, primary trait or task, sensor configuration, acquisition platform, and the main shift axes and methods addressed. These variables are summarized in Table 1, which provides a compact overview of the corpus and supports cross-study comparisons. The characterization is descriptive rather than statistical, reflecting the heterogeneity of crops, sensors, and tasks represented, but it provides a quantitative backbone for the thematic synthesis. It underpins, for example, the breakdown of robustness evidence across RGB and multisensor imaging (Section 4), the mapping between shift axes and acquisition or evaluation choices (Section 5), and the organization of adaptation strategies into a practical decision aid (Section 6).

Because the query targets specific major crops and explicitly filters for imaging and in-field or near-field platforms, the resulting corpus is naturally richer in cereals and orchards. Studies on related domains, such as root and soil imaging or greenhouse phenotyping, were retained in the core set when they satisfied the search criteria and contributed directly to the analysis of domain or distribution shift (for example, calibration transfer between hyperspectral platforms or cross-environment disease recognition). By contrast, works outside this scope are used only when they offer methodological insights and are explicitly treated as contextual rather than core evidence. This distinction between the 42-paper core corpus and additional contextual references ensures clarity about the evidential basis for the robustness claims synthesized in Sections 3–7 while still allowing methodological advances from the broader agricultural machine learning (ML) literature to inform the proposed taxonomy, protocols, and experimental agenda.

Table 1. Paper-level coding of the 42-study core corpus: crop/domain, task, sensing modality/platform, and the main shift axes and adaptation approaches emphasized in each study. Table 2 aggregates these codes into cross-study takeaways and actionable guidance.

Reference	Crop/Domain	Trait/Task	Sensors	Shift Axes & Adaptation Highlights
[36]	Root/soil imaging (minirhizotron; field and greenhouse)	Root segmentation under dataset shift	Minirhizotron root/soil imaging (proximal)	Cross-site + growth-stage (temporal) shifts; background/soil appearance and contrast/illumination differ across domains; segmentation degrades under shift.
[16]	Cereals (corn; UAV campaigns)	Leaf chlorophyll content (LCC) and average leaf angle (ALA) via PROSAIL-Net	UAV hyperspectral + PROSAIL simulations; solar-sensor geometry/multiangular views	Physics-guided sim→real transfer (PROSAIL pretraining + limited-label fine-tuning); domain gaps from geometry/illumination and protocol differences; multi-angular flights improve robustness.
[38]	Maize (field experiment)	Leaf chlorophyll content (LCC) with enhanced 2D-COS band selection + DNN/TL	UAV hyperspectral (field) + simulated hyperspectral (PROSAIL-like)	Within-season (growth stage/date) and treatment/cultivar variability; sim→field transfer learning with limited labels; sensitivity to acquisition timing motivates protocol-aware evaluation.
[14]	Grapevine orchards	Grape bunch detection/segmentation under domain shift	RGB (proximal/UAV-like viewpoints)	Scene (variety, canopy, background), viewpoint, and seasonal shifts; explicit domain-shift suite (leave-view/season/variety-out); fine-tuning and DA baselines.
[31]	Tomato (greenhouse/protected)	Leaf disease recognition under heterogeneous domain shift	RGB (controlled/greenhouse imaging)	Cross-domain disease classification; CycleGAN-like image translation and UDA; appearance shifts (lighting, background) and partial label space changes.
[8]	Wheat (multi-country field trials)	Head detection and counting (GWHD benchmark)	UAV RGB; proximal RGB	Large multi-site benchmark; scene and management shifts; explicit cross-site splits for OOD evaluation; protocol variations (GSD, altitude).
[39]	Apple orchards	Tree crown health-state detection (formed vs. unformed crowns) with YOLOv5; task-specific transfer learning	UAV RGB	Data-scarce, class-imbalanced setting; synthetic pretraining via object-based augmentation (OBA) using task objects/backgrounds improves over COCO initialization.
[21]	Field crops (soil moisture context)	Soil moisture estimation (OPTRAM + Deep Learning (DL))	UAV multispectral/hyperspectral (with OPTRAM indices)	Temporal and cross-field shifts in moisture; physics-informed indices + TL to stabilize water-related traits; protocol and soil-condition variability.
[11]	Potato (field trials)	Plant nitrogen content (PNC) using vegetation indices + hierarchical linear modeling (HLM)	Proximal hyperspectral reflectance (field spectrometer) + environmental variables	Cross-year and cross-region transfer; VI-PNC relationships vary with environment; hierarchical modeling with environmental covariates improves transferability over fixed/global fits.
[28]	Greenhouse crops (e.g., tomato, pepper)	Plant disease recognition with open-set domain adaptation	RGB (controlled imaging)	Cross-dataset and cross-environment shifts; closed-set vs. open-set UDA; pseudo-labeling and confidence filtering to handle unseen classes.
[12]	Row crops/weeds (e.g., soybean fields)	Weed/vegetation segmentation and mapping	UAV RGB/multi-spectral/thermal (depending on setup)	Cross-field and weather shifts (illumination, moisture); cross-platform or cross-altitude robustness; TL and protocol standardization; test-time normalization and calibration for management.
[5]	Winter wheat	Plant nitrogen content modeling; effect of spatial resolution on accuracy and cross-resolution transferability	UAV multispectral (native + resampled GSD; reflectance + VIs + texture)	GSD (scale) shift sensitivity; models degrade when applied across resolutions; CARS-selected features + RF to improve robustness across growth stages (2021–2022).
[6]	Soybean (yield prediction)	Yield prediction with spatial CV and ML optimization (RFE; RF/XGBoost/LASSO; stacked ensemble)	UAV multispectral VIs (+weather covariates)	Random vs. spatial-cluster vs. leave-one-field-out CV; independent field test for cross-field/year transferability.
[37]	Soil/biopore imaging across sites	Biopore detection and soil-structure characterization (DL segmentation/quantification)	Proximal RGB photography (excavated soil surfaces/soil-core cross-sections)	Multi-site, soil-type, tillage, and vegetation-history shifts; multi-site DL improves cross-site performance and robustness under varying soil appearance and noise.

Table 1. *Cont.*

Reference	Crop/Domain	Trait/Task	Sensors	Shift Axes & Adaptation Highlights
[7]	Apple orchards (MinneApple)	Fruit detection, instance segmentation, and counting (benchmark)	Ground-based RGB imagery (smartphone video frames; annotated instances)	Occlusion and illumination variability; train/test separated by different tree rows and different years to reduce leakage and better test generalization.
[40]	Rice (bacterial blight; disease severity)	Disease severity prediction with transfer learning (LSTM; transfer-ratio guidance; optional AT covariate)	Low-altitude UAV multispectral	Cross-condition and cross-sensor shifts (spectral/band differences); robustness under environment/instrument variation; few-target-sample transfer.
[32]	Sorghum (panicles) and wheat (heads)	Head/organ detection with CUT-based style-transfer domain adaptation (plus label refinement)	UAV/field RGB (unlabeled target domains for translation)	CUT image-to-image translation to bridge genotype, field condition, and lighting shifts; translated labels are corrected to handle missing/added small organs; improves cross-domain detection robustness.
[41]	Wheat	Above-ground biomass (AGB) estimation across growth stages (CBA-Wheat)	Proximal hyperspectral + UAV hyperspectral (VI-based)	Phenology-/thermal-time corrected VI→AGB model; cross-site and cross-season transfer; handheld→UAV evaluation to reduce bias under site/protocol differences.
[19]	Winter wheat and summer maize	Fuel moisture content (FMC)/canopy moisture status estimation (spectral indices + TL)	UAV multispectral + hyperspectral	Cross-crop transfer (maize ↔ wheat) with phenology/season variability; transfer learning to improve generalization; sensitivity to background/management differences.
[20]	Winter wheat	Plant N content estimation with multimodal fusion (RGB/MS/HS) and unequal-weight TL	UAV RGB, multispectral, hyperspectral	Cross-region and cross-sensor shifts; feature-level fusion with unequal weighting; TL across management regimes and sensors.
[42]	Citrus orchards	Chlorophyll estimation at multi-scale (leaf, canopy, UAV)	Proximal + UAV spectral sensing (multi-spectral/hyperspectral)	Scale shift (leaf↔canopy↔UAV); transfer learning across scales to mitigate data scarcity and improve cross-scale generalization.
[13]	Wheat	Ear counting (EarDensityNet) with ground→UAV transfer	Proximal RGB → UAV RGB	Cross-platform scale shift (altitude/GSD) and viewpoint differences; transfer learning from high-resolution ground images improves UAV counting robustness.
[35]	Wheat (breeding trials)	Grain yield prediction from UAV multispectral indices (NDRE1) with PLS; transfer across years and trial locations	UAV multispectral (red-edge/NIR)	Across-year and across-location generalization; strong dependence on measurement dates (growth stage timing); multi-date (date fusion) models often improve transferability.
[23]	Greenhouse HTPP facilities (maize)	Calibration transfer for trait prediction (RWC and nitrogen) across hyperspectral phenotyping systems	VNIR hyperspectral (multiple facilities/instruments)	Inter-/intra-facility spectral shifts under differing imaging conditions; spectral space adjustment via DS/PDS/DPDS/SST using transfer samples to reuse a master PLSR model without full recalibration.
[24]	Greenhouse HTPP/phenotyping systems	Common latent-space methods (TCA; di-PLSR) for hyperspectral calibration transfer	Hyperspectral (cross-system)	Latent-space harmonization across sensors/systems to reuse models under system-level acquisition differences; compares TCA-based alignment and domain-invariant PLSR variants.
[43]	Apple orchards (robotic harvesting)	Fruit semantic segmentation (sim→real)	Robot-mounted RGB-D (RealSense D435); synthetic orchard scenes (Gazebo)	Occlusion, lighting, and background variability in orchard scenes; synthetic data and domain adaptation to improve segmentation robustness under varied conditions.
[10]	Soybean (breeding trials; US ↔ Argentina)	Yield prediction from UAV multispectral imagery (transfer learning/fine-tuning)	UAV multispectral (cross-region)	Cross-site and cross-region generalization; transfer learning with limited target labels; sensitivity to temporal sampling (imaging schedule) in field trials.
[44]	Rice	Seedling detection in paddies	UAV RGB	Early-stage detection under varying water levels and background, cross-field GSD, and protocol shifts.
[4]	Rice (field plots; multi-year)	Grain-yield prediction with multi-temporal feature fusion + model updating	UAV RGB + multispectral (multi-date time series)	Growth-stage and cross-year drift (2017→2018); model updating with limited new-year labels; fusion of VIs with canopy structure to reduce background/senescence effects.

Table 1. Cont.

Reference	Crop/Domain	Trait/Task	Sensors	Shift Axes & Adaptation Highlights
[25]	Maize (regional suitability; soil constraints)	Soil water/salt content (SWC/SSC) inversion + water-salinity suitability mapping with ML + TL (TCA; UAV→satellite)	UAV multispectral + satellite multispectral	Cross-scale (UAV→satellite) transfer and regional variability; feature-space alignment via TCA; robustness to soil moisture/salinity domain shift.
[34]	Soybean (weed segmentation)	Weed semantic segmentation under weather/illumination shifts	UAV RGB	Weather-driven domain adaptation with CycleGAN-based translation; robustness across sunny vs. overcast conditions and across-year/mission appearance changes.
[45]	Soybean seeds (lab/controlled)	High-throughput seed phenotyping with synthetic data and Mask R-CNN	RGB (controlled imaging; synthetic + real)	Domain randomization/synthetic data for generalization across backgrounds and lighting; sim→real transfer; small-object detection robustness.
[17]	Soybean	Leaf chlorophyll content (LCC) estimation with HIT representation + transfer learning (Soybean-LCCNet)	Proximal canopy hyperspectral reflectance	RTM/PROSAIL-guided pretraining + HIT to stabilize hyperspectral inputs; robustness across growth stages and acquisition variability; TL improves LCC prediction over spectral-feature baselines.
[15]	Winter wheat	Leaf chlorophyll content (LCC) via PROSAIL-guided wavelength selection + transfer learning	Hyperspectral (ground and UAV)	Physics-guided sim→real: pretrain on PROSAIL simulations, then fine-tune on measured data; reduced wavelength set improves generalization and supports ground↔UAV transfer for LCC estimation.
[33]	Apple orchards	Large-scale orchard mapping with cross-sensor adaptation (UAV→Sentinel-2; Trans_GAN + TL)	UAV imagery + Sentinel-2 multispectral	Appearance and resolution gaps across platforms; image-to-image translation plus transfer learning to adapt segmentation from UAV labels to Sentinel-2 with limited target annotations.
[46]	Winter wheat	LAI/leaf chlorophyll content (LCC) and canopy chlorophyll content (CCC) via physics-informed transfer learning (PITL)	UAV multispectral + RTM/PROSAIL simulations	Physics-guided sim→real: RTM-informed pretraining/constraints to stabilize biochemical trait retrieval; evaluated across seasons (2022–2023) under field management/condition variability.
[47]	Winter wheat	Vertical (within-canopy) leaf chlorophyll content (LCC) profiling with RTM-guided deep transfer learning (LeafTNet)	Proximal multispectral leaf imaging	PROSPECT/RTM-sim pretraining → field fine-tuning; robustness across leaf layers and growth stages under varying N treatments over two seasons.
[26]	Winter wheat	Nitrogen nutrition index (NNI) estimation with multisensor UAV data and transferability analysis	UAV multispectral + RGB + thermal (TIR)	Cross-location and cross-cultivar transferability; feature selection/fusion and latent-space alignment (TCA) to reduce domain shift in N-status mapping.
[27]	Winter wheat (rust, scab, yellow dwarf)	UAV disease-area semantic segmentation with UDA (teacher–student + transformer)	UAV multispectral (RGB + 5 bands; DJI P4 Multispectral)	Cross-site and cross-year (2022–2023) shifts; site-held-out CV; robustness via pseudo-label UDA across locations/diseases.
[48]	Winter wheat	Plant nitrogen content (PNC) estimation; cross-management transferability	UAV multispectral (spectral + texture features)	8-year field experiment; transfer FP↔EI (management regime shift); fused spectral–textural features improve robustness across years/stages.
[18]	Maize	Above-ground biomass (AGB) via stem–leaf separation + 3D RTM + TL	UAV multispectral (with LESS/RTM)	3D vs. 1D RTM; cross-year and cross-field robustness; physics-guided architecture and TL; protocol and temporal shifts.
[9]	Rice	Grain-yield prediction with cross-sensor generalization (band-correction)	UAV multispectral (multisensor)	Cross-year and cross-cultivar transfer; explicit cross-sensor tests and band/spectral-response correction to reduce sensor-induced shift; robustness under stage/timing variability.

Table 2. Cross-study synthesis and empirical anchors for the four-axis shift taxonomy (scene, sensor, protocol, time): what typically changes, what breaks, what evaluation reveals it, and what practices most consistently help in proximal and UAV crop imaging.

Axis	What Changes in Practice	Common Failure Modes	Evaluation That Exposes the Gap	Robustness Levers (with Examples)
Scene	Cultivar/management, canopy structure, soil/background context, plot layout, geographic site	Spurious background cues; brittle texture/color heuristics; poor transfer to new farms	Leave-one-field/farm/site-out splits; cross-region validation (not random tiles)	Increase scene diversity; standardize framing; report per-site results; small target-set fine-tuning often recovers large fractions of lost accuracy (e.g., [8,14,31]).
Sensor	Camera replacement, bandset/spectral response, radiometry, optics, compression/pipeline differences	Spectral mismatch; calibration drift; domain gap, even with the same crop/field	Cross-sensor test on matched plots when possible; otherwise “sensor-out” within comparable protocols	Radiometric standardization; calibration transfer; feature alignment; explicit band sensitivity ablations (e.g., [5,9,23,24]).
Protocol	Altitude/GSD, view angle, time-of-day window, exposure/WB, panel usage, mosaicking, and preprocessing choices	Scale/viewpoint mismatch; illumination artifacts; inconsistent reflectance/color; hidden leakage from near-duplicates	Protocol ablations (GSD/window/view); protocol hold-out; ensure splits prevent correlated leakage	Lock exposure/WB where feasible; use panels/reflectance conversion; metadata logging; augmentation that matches protocol variation (e.g., [8,13,16,44]).
Time	Phenology/growth stage, inter-annual climate, rotations/management changes, disease progression	Seasonal drift; appearance evolution; label shift (class balance changes across season)	Season/year hold-out over the same field polygons; longitudinal splits	Phenology-aware planning; periodic recalibration; evaluate drift explicitly and report update frequency (e.g., [11,21,36]).

3. Robustness Under Distribution Shift: Definition and a Field-Focused Taxonomy

In proximal and UAV agricultural imaging, robustness under distribution shift means that a model’s accuracy and error structure remain within a small, predictable margin when conditions change in realistic ways. Four real-world factors are particularly relevant in this context (Figure 1): the scene, which includes canopy geometry, variety, netting, and background clutter; the sensor, which includes spectral response, optics, radiometry, and resolution; the acquisition protocol, which includes UAV flight altitude and timing, exposure and white balance, and sampling layout; and time, which includes phenology, season or year, and weather. To ground this taxonomy empirically, Table 2 summarizes recurring patterns observed across the 42-study core corpus, pairing each axis with representative failure modes, evaluation designs, and robustness levers (with concrete examples). In practice, robustness is assessed by testing models on independent fields, orchards, or seasons that differ along one or more of these axes, checking whether performance degrades smoothly rather than collapsing, and whether acceptable accuracy can be maintained without exhaustive relabeling or full retraining [7,8,10,35,36]. In plant pathology, continual-learning formulations provide a practical way to update models as new domains and categories arrive, while explicitly mitigating catastrophic forgetting [49]. This sets a clear deployment target in which models should hold up across farms and years, not only across random splits within a single field.

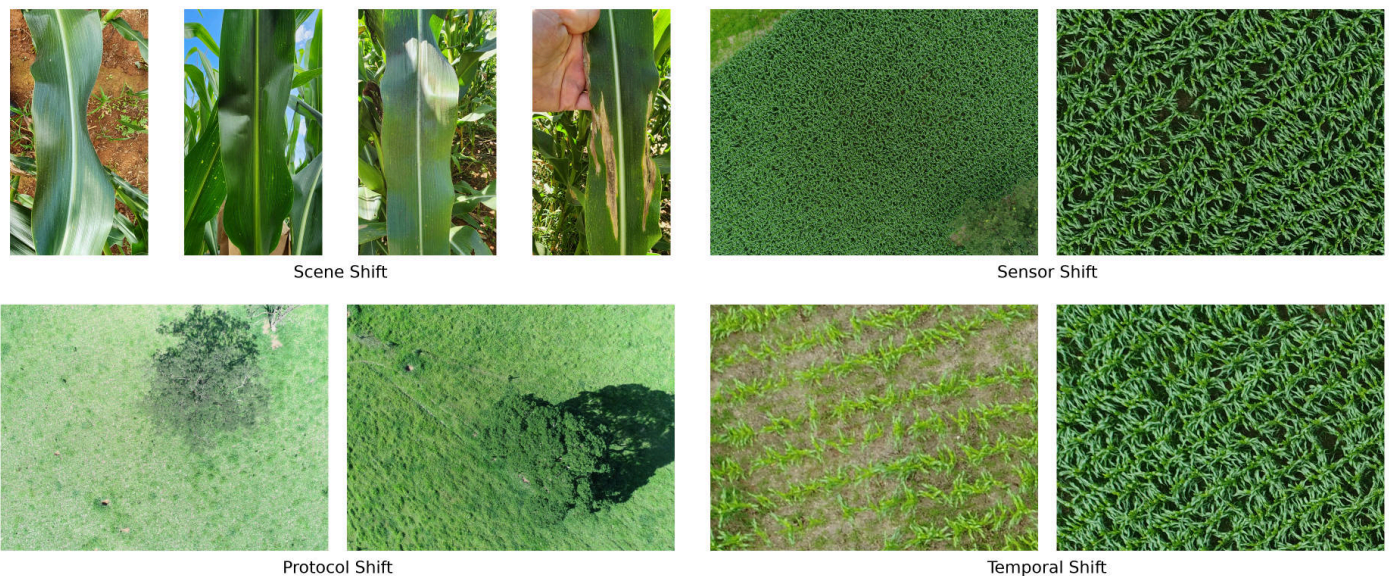


Figure 1. Representative sources of domain shift in agricultural RGB imaging (UAV and proximal). The *Scene shift* set (top-left, four examples) illustrates changes in the surrounding background and environmental cues (e.g., soil/ground texture, sky/vegetation backdrop, and the presence of an operator's hand). The *Sensor shift* pair (top-right) shows the effect of different imaging setups and effective resolution (different GSD or viewpoint) on the appearance of the same type of target. The *Protocol shift* pair (bottom-left) contrasts acquisition conditions under different lighting (near-noon vs. late-afternoon), affecting brightness, shadows, and color balance. The *Temporal shift* pair (bottom-right) shows crop appearance changes across growth stages, highlighting phenology-driven differences that can degrade model transfer if not accounted for. Together, these examples motivate treating robustness as axis-specific: some gaps can be mitigated by acquisition control (protocol), some by calibration/standardization (sensor), and some require explicit hold-outs and periodic updating (scene/time).

Conventional random cross-validation often mixes pixels or plants from the same rows and illumination pockets across the training and test sets, inflating transfer capability claims by leaking micro-context. This concern is especially well-documented in plant disease recognition, where limited dataset representativeness and capture-condition bias are repeatedly identified as primary barriers to field-ready deep learning systems [50,51]. This leakage is repeatedly documented in UAV yield-prediction and phenotyping studies, where random cross-validation overestimates transfer relative to spatial or field-level splits [6,35]. When evaluations instead hold out entire fields, orchards, or seasons, accuracy typically drops and error modes shift toward missed detections under occlusion, canopy shadows, or altered color and texture [8,35]. Fruit and ear-counting datasets corroborate this, as tests with independent orchards, varieties, or fields reveal failures that are missed under random splits, underscoring that independence between training and test sets in agriculture must be defined spatially and temporally, not only at the pixel or patch level [7,13,14].

Scene changes alter the distribution of geometries and textures. Leaf angles and densities, row spacing, under-canopy darkness, reflective soil or mulch, and the presence of nets or bags are all relevant factors in this context. For example, grape-bunch segmentation across varieties and viewpoints shows that models trained in one vineyard can become ineffective when bunch size, leaf glare, and occlusions change, even if the camera and protocol are fixed [14]. In apples, MinneApple and robotic-harvesting studies emphasize similar sensitivity to canopy structure, occlusion, and background clutter [7,43]. Category-level UAV benchmarks in cereals and weeds likewise report accuracy degradation when

management practices, canopy architecture, and field structure differ between training and evaluation sites, highlighting scene shift as a major source of error in counting, localization, and segmentation [8,12,13].

Changing cameras or lenses alters pixel statistics and feature salience, with factors like band placement, dynamic range, bit depth, lens distortion, and effective ground sampling distance (GSD) playing a crucial role. Hyperspectral calibration-transfer work has shown that even similar plant-phenotyping systems can require explicit spectral-space or latent-space mappings to maintain performance across instruments [23,24]. UAV nitrogen and yield studies further document that changes in sensor spectral response, dynamic range, or effective GSD cause systematic shifts in vegetation indices and texture features, affecting model transferability across devices and platforms [5,9]. These results underscore that sensor metadata and calibration are essential to reproducibility and to any credible claim of robustness, even when a similar RGB or multispectral setup is used.

UAV flight altitude and airspeed affect motion blur and effective GSD, while sun angle and cloud cover reshape shadows and color. In addition, sampling layouts (random tiles, row-following, and stratified designs) change the mix of backgrounds and phenological states. Empirical studies comparing different flight altitudes, overlaps, and acquisition timings show that holding GSD, sun elevation, and observation dates near phenological peaks often stabilizes performance across sites more than adding model complexity alone [5,35]. Field detection and segmentation studies also note that stabilizing motion, using sufficient overlap, and avoiding extreme insolation angles reduce false negatives in shaded or partially occluded targets, especially for small objects [12,13].

Phenology, season, and year introduce systematic appearance changes and alter clutter (weeds, residues, and soil moisture). Cross-season yield-prediction and nitrogen-content models in wheat, rice, and potato consistently report larger OOD gaps than within-season splits, so leave-one-season-out and independent-year tests provide the most unbiased estimate of deployable performance [4,6,9,11,35]. Similar temporal gaps are also observed in UAV disease mapping when models are evaluated on truly independent acquisition periods; for example, multi-site wheat disease segmentation datasets collected across consecutive seasons/years (e.g., 2022–2023) require explicit cross-site folds and highlight the difficulty of transferring across time and location [27]. Temporal shift further interacts with scene and protocol; the same altitude and exposure that work at heading may underperform at senescence due to color/contrast changes and increased occlusion from lodging and senescent leaves [4,35].

It is worth pointing out that shifts rarely occur in isolation. A farm adopting different trellising (scene) may mount a different camera (sensor) and fly at a new altitude (protocol) at a later phenological stage (time). Fruit datasets that take into account variety and viewpoint show that the angle of image capture can amplify occlusion under dense canopies and interact with pruning regime and canopy density [7,14]. Root phenotyping and soil structure imaging, as well as weed segmentation studies, similarly show that temporal shifts occur with cross-site differences in background and illumination, compounding domain shift for segmentation models [34,36,37]. Wheat head and ear counting studies reveal that small resolution drifts and changes in observation distance worsen head separation precisely when phenology changes head shape and contrast [5,8,13]. Treating the four axes explicitly helps plan acquisitions (which combinations to cover) and decide which adaptation strategies to prioritize first in each deployment.

To make robustness claims comparable, evaluations should include at least one independent field or independent season test aligned with the intended deployment gap [4,6,9,35]. They should specify sensor details (bands, bit depth, lens, nominal GSD, and radiometric treatment) and protocol settings (altitude, overlap, exposure and white-balance

rules, and solar elevation range) and disclose the sampling layout and spatial decorrelation strategy used during training and validation [5]. Fruit and orchard studies benefit from reporting canopy structure, netting, and pruning regime, as these consistently affect failure rates under transfer [7,14]. UAV weed and disease benchmarks additionally recommend enumerating farm-level factors (soil background, inter-row cover, irrigation, and management regimes) that introduce variability, both to avoid spurious shortcuts and to correctly interpret cross-site gaps [12,27]. Global wheat head datasets similarly document field-level factors such as cultivar, management, and trial site [8].

A practical view is to define robustness thresholds across the four axes, that is, how much GSD drift, exposure variance, variety change, or phenological gap the model can tolerate before recalibration or fine-tuning is needed. Independent field and season experiments make this explicit by plotting accuracy or error against controlled deviations in altitude, year, or viewpoint, or against different cross-validation schemes [5,6,12,35,36]. Simple sanity checks and illumination diagnostics, per-field resolution checks, basic spectral calibration comparisons, and occlusion profiling help distinguish sensor and protocol issues from scene and time effects before resorting to more complex adaptation or retraining [23].

The taxonomy thus guides both evidence synthesis and method selection. In crops and orchards, scene and time shifts dominate counting and detection errors, so evaluation should employ leave-one-field-out and independent season tests, as done in wheat head, ear counting, fruit, and weed benchmarks [7,8,12,14,27]. In addition, sensor and protocol standardization or explicit calibration-transfer procedures should be reported alongside accuracy to clarify whether performance is more related to the model architecture or to the acquisition and preprocessing pipeline [5,9,23]. The following sections address which adaptations (normalization and augmentation, small target-set fine-tuning, feature alignment, or style translation) are the most effective for each type of shift, with benchmark datasets that stage variety, viewpoint, site, sensor, and season changes providing appropriate reference points to compare remedies across tasks [10,12,27,36].

4. Empirical Evidence Across RGB and Multisensor Imaging

Robustness under distribution shift depends as much on what is measured as on how and where it is measured. Across the reviewed studies, RGB emerges as the practical baseline for in-field sensing because of its ubiquity, cost, and relative ease of standardization. When acquisition protocols are controlled, and evaluation is spatially aware, RGB models already transfer better than expected in several tasks, particularly object-level detection (heads, fruit, organs) and canopy-level counting or segmentation in daylight conditions [7,12,13,32]. In orchards, detectors trained on RGB imagery with block-wise or row-wise splits generalize across rows and blocks more reliably than models trained and evaluated with random patch splits, which tend to inflate accuracy and mask sensitivity to background, cultivar, and illumination changes [7,43]. For field crops, RGB works well when flight height, ground sampling distance (GSD), and view geometry are held consistent, as shown in wheat head, ear counting, and weed segmentation studies that emphasize careful protocol design and spatially structured validation [8,13]. However, performance degrades under oblique views, long shadows, high residue, or different sowing densities, unless the training set or augmentation strategy explicitly covers these shifts, as documented in cereal head and seedling detection and synthetic data soybean seed phenotyping [12,44].

The empirical evidence favors multispectral, hyperspectral, and thermal modalities in regimes where RGB-only cues are less stable under shift—namely when the target signal is genuinely spectral or physiological (for example, stress, nutrient status, chlorophyll content, leaf water content, or canopy temperature) or when lighting and background are unstable

and spectral features provide information that RGB lacks. In the subset of studies that report explicit modality ablations or paired baselines, adding red-edge/NIR (and, where available, thermal) consistently reduced cross-date and cross-field degradation for vigor-, chlorophyll-, and nitrogen-related traits compared with RGB-only baselines, especially when soil brightness, residue, and illumination differ across sites and seasons [5,9,10,19]. In field crops, red-edge and near-infrared (NIR) bands systematically serve as nutrient and vigor proxies across dates and sun angles, improving transfer to new parcels and years with different soil brightness, residue levels, or management practices, for both handcrafted indices and learned representations [4,5,9,10,19]. Hyperspectral and multispectral approaches that couple radiative transfer models with deep networks or recurrent architectures demonstrate that narrowband cues can maintain chlorophyll, nitrogen, and biomass estimation accuracy under significant changes in acquisition date, geometry, and site for wheat, soybean, maize, and rice [15–18]. In dense canopies, paddies, and protected environments, such narrowband or physics-guided cues mitigate saturation and mixed-pixel issues that challenge RGB classifiers, especially for early stress detection and subtle disease symptoms where color and texture changes are weak [23]. Thermal and water-content-oriented imagery adds robustness for water-related traits under midday illumination changes by anchoring canopy temperature or moisture indices relative to air and soil conditions, although models still benefit from co-registered RGB or multispectral imagery to provide structure and context for segmentation and detection [17,27].

At the same time, extra bands do not automatically improve robustness. Several studies show that simply adding multispectral, hyperspectral, or thermal channels can degrade transfer if radiometric handling, band selection, or cross-sensor alignment is weak. Under domain shift, noise and sensor-specific artifacts quickly propagate through deep models, and redundant or unstable bands make models more sensitive to hardware and protocol changes [5,9,23,24]. As a result, a smaller set of more informative bands often outperforms the full spectral set, especially when combined with band selection, spectral compression, or physics-guided feature construction [15,46]. Practical gains arise when pipelines compress or regularize spectra to reduce sensor-specific idiosyncrasies, harmonize exposure, white balance, and irradiance across acquisition sessions, and fuse modalities at the feature level with explicit normalization rather than raw concatenation of channels [9,23,24]. In orchards, multispectral and hyperspectral data are most helpful for maturity, quality estimation, and subtle disease or health signals, whereas counting and detection tasks usually employ RGB images unless illumination is highly variable or fruit color closely matches the background [7,39]. In those cases, adding a red-edge band or simple vegetation indices can improve model transfer without major architectural changes [9,20].

Across tasks and crop types, consistent patterns emerge. For counting and detection problems, RGB with strong acquisition control and spatially aware evaluation is typically the most efficient choice. Multisensor gains appear mainly when canopy texture and target color overlap (for example, green fruit against foliage), when severe specular reflections and shadows are recurrent across sites, or when class boundaries are subtly spectral rather than purely geometric [7,12]. For stress, nutrient, and disease grading, multispectral, hyperspectral, and thermal modalities more consistently improve robustness, provided band handling and calibration are explicit and stable over time [5,26,46]. Disease and stress studies under greenhouse or semi-controlled environments also show that spectral cues tend to be disproportionately useful because artificial lighting and engineered backgrounds create shifts that spectral normalization and calibration-transfer methods can dampen [23,24]. Even in these settings, however, geometry control (camera distance and angle) remains critical for transfer, as reflection effects and occlusion patterns still cause errors [32]. Models

usually generalize better across seasons when they exploit multi-date context or temporal smoothing, since aggregating predictions over several acquisition sessions makes them less sensitive to sporadic lighting or management anomalies [4,10,35].

Cross-platform and cross-sensor experiments further characterize the role of additional modalities. For UAV-to-UAV transfer (different cameras or lenses), models trained with explicit radiometric standardization and sensor alignment perform better than those relying on raw reflectance estimates alone. Even simple per-band scaling based on reference panels or pseudo-invariant features reduces drop-off on new hardware and between different multispectral cameras [5,9,23]. For UAV-to-proximal and proximal-to-UAV transfer, domain gaps in view geometry and reflectance-distribution effects dominate. Studies on wheat yield, biomass, and nitrogen status show that combining shallow structure cues from RGB with a few stable spectral features often outperforms either modality alone, particularly when UAV flight altitudes vary across sites or when canopy architecture differs between ground and aerial views [4,26]. Thermal transfer improves when canopy temperature is normalized by ambient conditions, wind, and reference surfaces, not just absolute temperatures, highlighting that protocol metadata (such as air temperature and wind speed) are relevant and must be logged to ensure reproducible robustness [12,26]. Physics-guided sim-to-real strategies that pretrain on radiative-transfer simulations and then adapt to real data also help bridge sensor and platform gaps for chlorophyll and biomass traits [15–18].

In summary, the empirical evidence suggests four practical guidelines for choosing modalities under a distribution shift. First, start from a robust RGB baseline evaluated under spatially aware splits, thus establishing a realistic reference for transferability [7,8,13,35]. Second, add red-edge or NIR when the trait of interest is physiological or when illumination and soil/background conditions vary substantially over time or across fields, and consider hyperspectral or physics-guided spectral features when chlorophyll and nitrogen are primary targets [5,9,15,17,46]. Third, consider thermal for water stress assessment, irrigation timing, and heat-related risks, but normalize canopy temperature by ambient conditions and wind, and combine it with cues extracted from RGB or multispectral images [12,26]. Fourth, prioritize band handling, alignment, and protocol standardization before increasing spectral depth, because under domain shift, how data from different flight missions and sensors are made comparable often matters more than how many bands are available [5,9,23,24]. These patterns hold across field crops, orchards, and controlled environments in the reviewed studies, and they connect directly to the acquisition and evaluation choices discussed in the following sections.

5. Acquisition, Standardization, and Evaluation Under Distribution Shift

Robustness in proximal and UAV imaging starts with how data are acquired and standardized. Many of the gaps observed in cross-field and cross-season tests can be traced to choices made before any model is trained [35]. This section consolidates practical guidance from the reviewed studies into a coherent workflow, from flight planning and camera configuration to radiometric handling and evaluation design.

As a guiding framework, Figure 2 summarizes how shifts along the four axes (scene, sensor, protocol, time) translate into observable effects in model behavior and how these symptoms can be addressed through a graduated set of remedies, from low-cost normalization and augmentation to more involved domain-adaptation and test-time strategies.

Acquisition should be planned around agronomic phases that maximize target visibility while minimizing confounders. Good moments for acquisition include when heads are visible and not yet lodged for cereals, when bunches and fruits are not yet subject to heavy occlusion or bagging in orchards, and when canopies are dry and reasonably stable under wind [7,8]. Many studies recommend constraining flights to a solar window (for example,

late morning to early afternoon) or, when this is not possible, deliberately sampling multiple times of the day and documenting them, because sun elevation and cloud cover have a strong influence on shadows, specular reflections, and apparent color [4]. Relatedly, in multi-temporal UAV campaigns, logging sky conditions (haze, aerosol, variable overcast) and filtering acquisitions that violate illumination constraints helps stabilize sequential inputs across dates [52]. Maintaining a roughly constant ground sampling distance (GSD) across fields and dates also emerges as a simple but powerful control. When this is not feasible, acquiring at two nearby altitudes and reporting model sensitivity to different GSDs (such as 1–2 cm versus 3–5 cm) helps quantify how resolution interacts with domain shift [12]. Recording cloud cover, wind, and leaf wetness is equally important, since wind increases motion blur and parallax errors, while wet foliage changes reflectance and temperature patterns, and soil moisture and salinity strongly modulate spectral and thermal responses in water-limited environments [12,21].

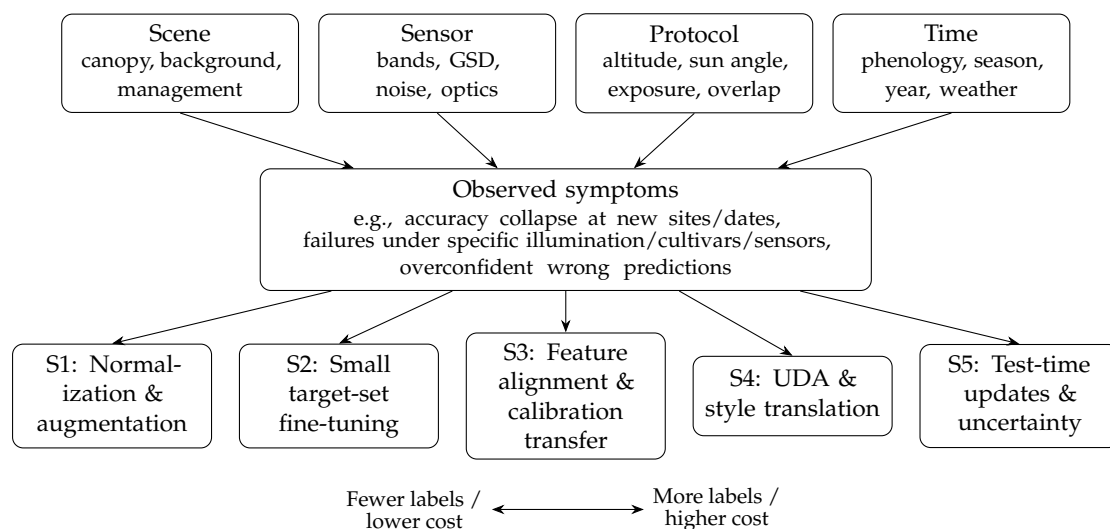


Figure 2. Roadmap from shift diagnosis to deployable remedies in proximal and UAV crop imaging. Top: the four shift axes (scene, sensor, protocol, and time) organize the space of failure causes. Middle: observable symptoms (e.g., site-to-site collapse, seasonal drift, protocol sensitivity) help diagnose which axis dominates. Bottom: a staged response from low-cost measures (normalization/augmentation) to more involved updates (small target-set fine-tuning, calibration transfer/feature alignment, UDA/style translation, and test-time updates). The intent is not to prescribe a single method but to match interventions to the dominant gap under field constraints (labels, compute, and acquisition control).

For RGB imaging, locking exposure, ISO, and white balance and, where possible, recording RAW images reduces variability due to in-camera tone mapping and automatic gain control [35]. Calibrating lens distortion and vignetting and checking focus at the operating altitude are basic but often ignored steps. For multispectral and hyperspectral sensors, good practice includes capturing dark frames, sensor temperature, and spectral response metadata. For thermal cameras, this involves recording emissivity settings, ambient air temperature, and wind information, ideally with at least one reference object of known emissivity when materials differ in the scene [26]. View geometry should be kept consistent (for example, predominantly nadir flights with limited roll and pitch), or, if off-nadir views are intentional, they should be sampled in a controlled way, and performance according to view angle should be reported [44].

Several works show that radiometric treatment is central to transferability. Panel-based normalization, or the use of pseudo-invariant features (PIFs), reduces cross-date and cross-camera drift [53,54] more reliably than histogram matching alone [5]. For multispec-

tral and hyperspectral sensors, conversion to reflectance using an incident light sensor is preferred when available. Otherwise, panel-based scaling should be applied and residual variability documented [16]. When transferring between cameras, aligning bands by their spectral response functions and, in some cases, learning a compact band space, such as Principal Component Analysis (PCA) or latent-space representation, helps reduce sensitivity to sensor-specific characteristics [23,24]. Recent multisensor nitrogen and Nitrogen Nutrition Index (NNI) studies further illustrate how such calibration-transfer and latent-space harmonization steps improve cross-camera robustness [26]. For thermal imaging, normalizing canopy temperature by ambient air temperature and wind (for example, using $\Delta T = T_c - T_a$ or similar indices) improves stability across microclimates and seasons and is a recurring recommendation in irrigation and stress-monitoring studies [26].

Geometric alignment matters both for mosaics and for multimodal fusion. High forward and side overlap (on the order of 70–90 % in each direction), combined with RTK/PPK or well-distributed ground control points, reduces between-mosaic warping that can otherwise cause label noise and degrade performance when tiles from the same field are split across folds [9,16]. For proximal imaging, fixing standoff distance and view angle with physical objects limits BRDF-related variation and makes repeated scans more comparable [16]. When fusing RGB with multispectral, hyperspectral, or thermal data, extrinsic calibration and per-band alignment should be verified along salient edges before feature-level fusion is attempted. Several studies report that misregistration, rather than model choice, dominates errors in naïve multimodal pipelines [26].

The reviewed evidence favors simple, auditable preprocessing controls that generalize, including locking white balance and exposure where feasible, applying panel- or PIF-based linear scaling, using light dehazing or color-constancy corrections only when acquisition conditions justify them, and adopting band selection guided by agronomic priors or physics-based simulations rather than unconstrained channel concatenation [5,9,16]. Any transform that could entangle training and test domains (e.g., tiling strategy, CLAHE, denoising) should be documented and kept identical across splits [6]. Finally, core metadata should be logged, since even lightweight models using a small set of metadata cues can generalize better across sites and seasons than more complex models that ignore these factors [10,35].

Acquisition and standardization alone are not sufficient, and evaluation must be structured so that robustness claims truly reflect shifted distribution behavior. A recurrent recommendation is to treat the field block, management unit, or farm as the fundamental independence unit and never tile a single mosaic into both training and test folds [7,8,35]. For detection and segmentation, image footprints should not cross split boundaries, and all dates from a given field or block should stay within the same split to avoid temporal leakage. Most studies that explicitly target robustness report three families of splits: an in-domain (ID) random-by-unit split within a site as a sanity baseline; a spatially structured cross-validation (for example, block-wise or leave-one-field-out) to quantify within-region transfer; and at least one leave-one-site, leave-one-season, or leave-one-sensor-out configuration that matches the intended deployment gap and serves as the main robustness indicator (Figure 3) [6,9,35].

To make these trends more concrete, how often each out-of-distribution (OOD) evaluation type appears in the 42-study core corpus (as coded in Table 1) is summarized as follows: at least 16/42 studies include an explicit cross-field/site split; 10/42 include a cross-season/year hold-out; 4/42 include a cross-sensor test; and 8/42 report protocol stress tests (e.g., altitude/GSD, view angle, acquisition window, or exposure policy). These tallies are conservative because many papers describe the shift qualitatively or mix several axes without releasing split files or reporting matched in-domain (ID) and OOD scores. Where paired ID and OOD metrics are available, it is recommended that both the absolute OOD

score and the drop under shift (e.g., Δ macro-F1, Δ mIoU, or Δ mAP in percentage points) are reported because the drop communicates shift severity more directly than the ID score alone and enables clearer cross-paper comparisons even when tasks and metrics differ.

For detection and instance segmentation tasks, mAP@[0.5:0.95] and class-wise AP are standard; for semantic segmentation, mean IoU and boundary-aware scores; for classification, macro-F1 and per-class recall; and for regression, MAE, RMSE, and R^2 . Robust studies complement these with 95 % confidence intervals via block- or field-level bootstrapping or non-parametric paired tests across units, emphasizing effect sizes rather than only p -values [6]. Calibration metrics such as expected calibration error or Brier score [55] are recommended when outputs feed thresholded actions like spraying or irrigation, and some works additionally stress reporting error distributions by management unit rather than only global averages [12,26]. Breaking results down by shift axes helps reveal where performance collapses and guides targeted augmentation or adjustments in acquisition protocols [7].

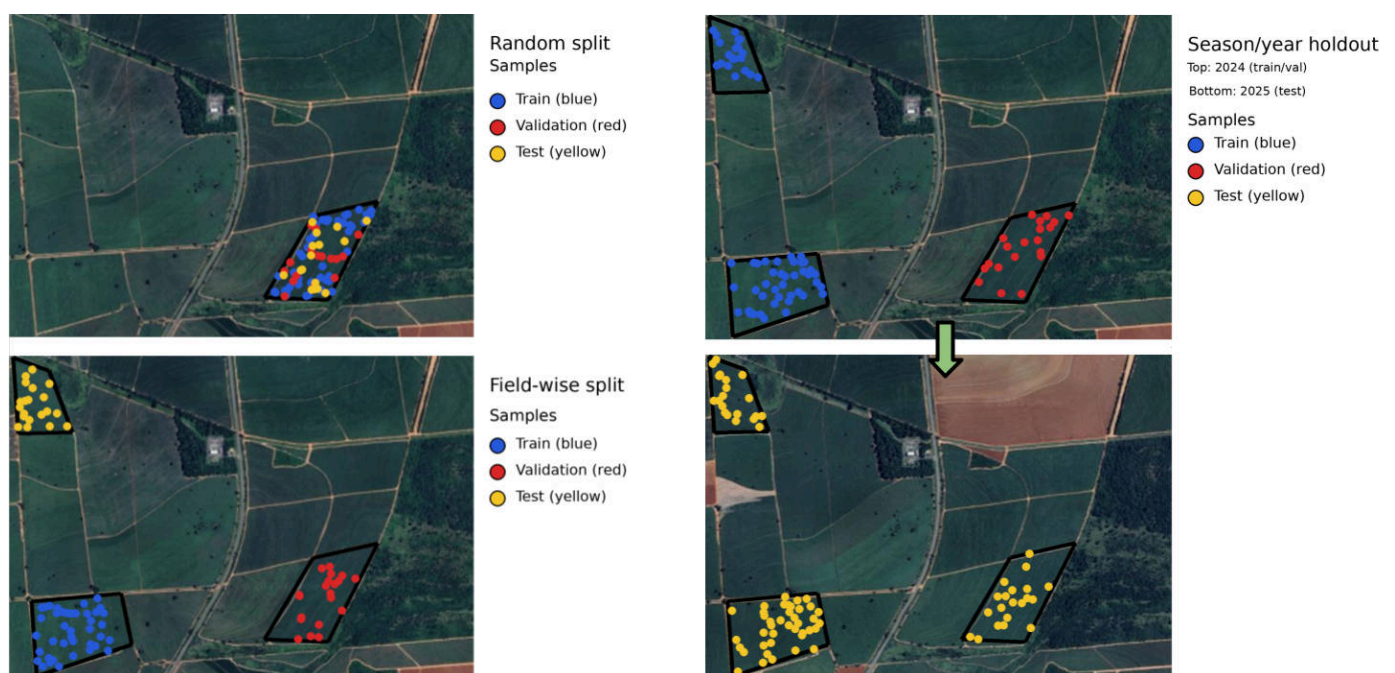


Figure 3. Illustrative training, validation, and test splitting strategies for crop imagery (blue = training, red = validation, yellow = test). Top-left (random split): samples from a single field are assigned across all subsets, which can introduce spatially correlated leakage and over-optimistic performance. Bottom-left (field-wise split): entire road-bounded fields are assigned to a single subset, reducing within-field dependence between splits. Right (season/year hold-out): training/validation uses one year (top: 2024) while testing uses another (bottom: 2025) over the same field polygons, approximating deployment under temporal (seasonal/inter-annual) shift. As a reporting minimum for robustness, we recommend including at least one field-wise (or farm-wise) split and one season/year hold-out split; random splits can be reported only as an in-distribution reference because they are prone to spatial/temporal leakage.

Finally, several articles highlight the importance of ablation studies that explicitly test robustness levers. Acquisition ablations vary solar window, GSD, and view angle to show the benefit of standardized protocols [12]. Preprocessing ablations compare pipelines with and without panel-based normalization, or full spectral stacks versus compressed band sets and calibration-transfer transformations [5,23,24]. Modality ablations contrast RGB baselines with incremental additions of red-edge/NIR and thermal channels and feature-level versus early fusion [5,26]. Adaptation ablations compare no update, small target-set fine-tuning, feature alignment, and unsupervised domain adaptation or style translation

on the same out-of-distribution split [27,28]. A well-documented study reports ID, spatial cross-validation, and at least one out-of-distribution result, demonstrates that acquisition and standardization choices reduce generalization gaps, quantifies uncertainty at the unit level, and releases, where licensing permits, sensor specifications, flight parameters, panel descriptions, preprocessing code with fixed seeds, and exact split strategies so that shift-aware findings can be reproduced and extended [5,6,35].

6. Adaptation and Updating Under Distribution Shift: A Practical Decision Aid

Robustness in real deployments rarely comes from a single technique. Instead, it depends on matching specific shift symptoms to lightweight solutions that respect field constraints such as limited labels, tight compute, and changing illumination and canopy structure. This section consolidates adaptation options, from simple normalizations to unsupervised domain adaptation, into deployability-oriented guidelines for proximal and UAV imaging across RGB, multispectral, hyperspectral, and thermal data. Many first-order robustness gains come from the acquisition and preprocessing controls summarized in Section 5 (e.g., exposure/WB policy, radiometric scaling, and protocol standardization). Here, the focus is on model-side updating strategies that remain after those controls are in place, organized by label availability and by the dominant shift axis.

When a modest fraction of the target site can be labeled, typically on the order of 10–30 %, small target-set fine-tuning becomes a powerful next step. In practice, this often means retraining the task head while freezing most of the backbone, followed by cautious fine-tuning of upper layers with discriminative learning rates so that early features remain generic rather than adapted to local quirks. Class-balanced and difficulty-aware sampling, deliberately including occluded, overexposed, or otherwise challenging examples, allows the small labeled set to “teach” the critical gaps. Early stopping on a spatially independent validation block, instead of random tiles, prevents excessive optimism caused by leakage and preserves a realistic view of transferability [7,35]. This strategy is effective for RGB baselines in disease, fruit, and head detection [7,8,14], as well as for multispectral and hyperspectral trait mapping when spectral shifts are moderate [15,17,26]. Several multi-year yield and nitrogen studies show that such targeted fine-tuning on a few representative fields can close a large fraction of the cross-season gap [4,6,35].

When labels in the target domain are scarce or unavailable, feature-alignment techniques based on shallow statistics offer label-free or label-light options. Classical color and histogram matching, applied per field and after panel-based scaling, can bring RGB mosaics from different missions into closer correspondence, provided that overcorrection is avoided [35]. In feature space, alignment approaches such as transfer component analysis (TCA) equalize means and covariances of feature distributions between source and target domains and can be inserted as lightweight modules or offline transforms. They pair well with small target label sets and provide quick robustness gains [20,23,24]. For multispectral and hyperspectral data, band transforms that map sensors into a common latent basis help decouple model behavior from the particularities of individual cameras [9]. These methods are computationally inexpensive, auditable, and suitable for on-farm updates where resources are limited.

For larger domain gaps under zero or very limited labeling capabilities, unsupervised domain adaptation (UDA) becomes attractive. Mean-teacher or exponential-moving-average consistency schemes enforce prediction stability under realistic augmentations on target images, with a teacher network smoothing updates and reducing overfitting. Related consistency and domain-adaptation strategies have been applied to multisensor chlorophyll and nitrogen mapping across seasons and sites [27]. Adversarial feature align-

ment, where a domain discriminator pushes features toward domain invariance [56], can further reduce appearance-related discrepancies, although care must be taken to avoid class-conditional misalignment (for example, confusing leaves with background under new lighting conditions). Pseudo-labeling with confidence filtering, sometimes combined with spatial priors such as row geometry, can progressively label easy target examples and increase effective supervision. Open-set and closed-set UDA variants for plant disease recognition in greenhouses illustrate this progression [28,31]. Overall, UDA is most useful when acquisition differences dominate the gap (lighting, soil background, sky conditions, sensor characteristics) and class taxonomies are the same across domains [27,28].

When inter-sensor or weather effects dominate pixel statistics, style or appearance translation can be used to bring source images closer to the target “look”. CycleGAN, CUT, and related frameworks translate source imagery into the target style (illumination, color, texture) without target labels, after which standard training or fine-tuning is performed on these translated pairs [31–34]. Content-preserving constraints that emphasize edges and structure are critical in yield-relevant applications so that objects such as panicles, bunches, and fruits are not hallucinated or distorted [32]. Style translation is particularly useful when color and contrast shifts are the main failure mode (for example, weather-induced appearance changes or cross-camera color differences). It is less appropriate when structural differences, such as canopy architecture or trellis type, are the dominant source of variation [12,36].

Several techniques operate directly at test time to provide plug-in robustness during deployment. Batch-normalization recalibration, where statistics are updated per field or per mission, can adapt feature distributions without labels. This can be combined with entropy minimization on target inputs to sharpen predictions [57,58]. Normalization schemes that explicitly condition on metadata such as solar elevation, GSD, or temperature (for example, through conditional batch normalization or metadata-conditioned heads) further reduce shift by tailoring affine parameters or decision thresholds to acquisition conditions [26]. These methods are fast and label-free but must be monitored for drift, with a clean rollback path to a stable baseline model.

Because labels are costly in the field, active learning and data selection strategies can make each annotation count [59]. Query strategies that target spatial novelty and failure sources (regions of low confidence, high entropy, or uncommon backgrounds) help select the most informative patches for labeling. Selecting entire blocks or representative tiles across field heterogeneity, rather than random tiles, improves coverage of key conditions with fewer labels [6,35]. Some studies implicitly follow this logic when constructing multi-year training sets or when adding targeted labels on crops, cultivars, or management regimes that initially show the largest generalization gaps [10]. While formal data-valuation methods remain rare in proximal and UAV phenotyping, their underlying principle, identifying redundant imagery and highlighting rare, high-leverage situations, aligns well with these practices.

Uncertainty estimation and calibration add another layer of robustness. Shallow ensembles, using just two or three independently trained seeds or backbones, often deliver observable cross-site gains with modest computational overhead and provide dispersion estimates that highlight failure modes [6]. Simple post-hoc calibration, such as temperature scaling on a spatially independent validation block, improves the reliability of confidence scores and thus the thresholds used for actions like spraying or harvesting, especially when shifts alter the distribution of output scores. In irrigation, stress mapping, and nitrogen management contexts, calibration is particularly important because small changes in score distributions can translate into discrete management decisions and costly interventions [55].

In practical deployments, it is useful to think in terms of a symptom–remedy correspondence rather than a fixed recipe. If a model performs well in controlled tests but fails under harsh sun or overcast conditions, the first line of response is to strengthen illumination and color augmentation, enforce panel or PIF normalization, and consider batch-normalization recalibration or entropy-based test time adaptation. If color statistics remain the main issue, style translation may be justified [12,32]. When moving to a new site with different soil, residue, or background, shallow feature alignment (e.g., TCA) combined with small target-set fine-tuning, and possibly geometry-aware priors, tends to provide quick improvements [23,26]. For sensor changes, such as a new UAV camera with a different spectral response, spectrally aware band mapping, or compact band spaces, combined with minimal fine-tuning, can mitigate mismatch. If thermal imaging is involved, normalizing canopy–air temperature differences by ambient conditions and wind is essential [26]. Seasonal and phenological drift often requires curated target sets that consider key stages, augmented by teacher–student UDA for unlabeled data and geometry-aware augmentations or modest ensembling [35]. When the deployment shift is driven by evolving class sets (e.g., new varieties/market categories), adaptive class-incremental updating combined with domain adaptation can reduce the need for full retraining while controlling forgetting [60]. When dealing with sparse, rare classes (early disease symptoms, small fruit), active learning focused on uncertain or rare patches and class-balanced fine-tuning are generally preferable to heavy UDA, which can dilute minority signals [59].

A recurring caveat across these adaptation families is that robustness gains are not monotonic. Negative transfer can occur when class priors change, when the target domain contains unseen sub-classes, or when alignment enforces invariances that erase discriminative cues [28,31,56]. Pseudo-labeling and teacher–student UDA are particularly sensitive to confirmation bias under phenological mismatch and under sensor noise or radiometric instability, which can cause drift if updates are not constrained and monitored [6,35]. Likewise, style translation can inadvertently “fix” a domain gap by altering agronomically meaningful structure (geometry/texture), so content-preservation checks and conservative use are essential [32]. Finally, test-time adaptation can silently degrade without safeguards, so this should be treated as a monitored, reversible component (with stable baselines, calibration checks, and truly held-out field/season tests) rather than a substitute for shift-aware evaluation [6,26,58].

In practical deployments, it is useful to think in terms of a symptom–remedy correspondence rather than a fixed recipe. When performance drops are dominated by appearance/protocol changes, prioritize augmentation and lightweight updates (e.g., normalization or test-time adaptation) before heavier pipelines [6,26]. When the dominant gap is a scene or site change, small target-set fine-tuning and shallow alignment are often the fastest reliable options. When sensor mismatch is primary, spectrally aware band mapping (and, if needed, limited fine-tuning) should be used before introducing complex UDA [31,56]. For seasonal drift, measures include curating small target sets spanning key stages and comparing lightweight updating against UDA under the same OOD split [61]. Table 3 summarizes these trade-offs (label needs, auditability, and negative-transfer risks) as a deployment-oriented decision aid.

Cost, risk, and governance considerations are relevant across all these options. Many gains come from normalization, augmentation, and a few labeled blocks. Heavier UDA and style translation should be reserved for clearly documented appearance gaps. To avoid “silent” target leakage, independent fields must be preserved for final testing, and no hyperparameter should be tuned on the held-out site [8,35]. Adaptation steps should be documented explicitly, including which layers were fine-tuned, how batch-normalization statistics were updated, and which alignment or translation transforms were applied,

along with the data used. Maintaining a stable baseline model for rollback and auditing is essential in operational settings [6,26].

While the literature surveyed here is heterogeneous in terms of crops, tasks, and protocols, a few comparative patterns recur across proximal and UAV settings. First, *label requirement* is the main practical divider: small, diverse target sets enable fast recovery via fine-tuning, whereas UDA/translation approaches target zero- or low-label settings but introduce additional failure modes. Second, *auditability and governance* differ markedly: lightweight updates (normalization, small-set fine-tuning, shallow alignment) are typically easier to document and roll back than heavier UDA or image-to-image translation pipelines. Third, methods vary in *negative-transfer risk*: alignment or UDA can harm performance when class priors shift, rare classes matter, or class-conditional structure changes, whereas calibration and ensembling mainly affect decision reliability and do not correct systematic bias. Finally, appearance translation can be effective for large pixel-statistics gaps, but it must preserve geometry and agronomically meaningful structure to avoid “fixing” the domain gap by distorting the signal of interest. Table 3 summarizes these trade-offs as a decision aid that complements the staged roadmap in Figure 2.

Table 3. Comparative critique of common robustness/transferability method families in proximal and UAV crop imaging. The goal is not to prescribe a single best approach but to make trade-offs explicit (labels, complexity, auditability, and failure modes) so results can be interpreted and compared under realistic OOD splits.

Method Family	Target Labels	Best-Matched Axes	What It Typically Improves (When It Works)	Main Failure Modes/Cautions (Critical)
Small-set fine-tuning	Low (few blocks)	Scene, time (moderate protocol) Fast recovery on new sites/seasons by re-anchoring higher-level features when the labeled target blocks span key backgrounds/phenology [4,6,14,35].	Overfitting when target labels are narrow/spatially correlated can yield optimistic estimates if tuning is performed on the deployment domain without a truly held-out field/season.	
Shallow feature alignment (e.g., TCA-style; calibration transfer)	None–low	Sensor, scene (some protocol)	Stabilizes representations across devices/platforms as an auditable, lightweight module or offline transform; useful when sensor mismatch dominates [9,23,24].	May misalign class-conditional structure (hurting rare classes) and underperform when the main gap is phenology/time drift rather than sensor statistics.
Adversarial alignment (DANN-style)	None–low	Protocol/appearance-heavy gaps	Reduces global domain discrepancy without target labels when acquisition differences dominate [56].	Higher negative-transfer risk under label shift or class-conditional mismatch; may “wash out” discriminative cues if invariance is over-enforced.
Teacher–student/consistency UDA (pseudo-labeling)	None (or very low)	Scene, protocol, time (with streams)	Can improve robustness using unlabeled target data under controlled augmentation/consistency when the label space is shared across domains [27,28,31].	Confirmation bias/pseudo-label drift; sensitive to unseen classes or evolving distributions; requires conservative thresholds and monitoring.
Style/appearance translation (CycleGAN/CUT-like)	None	Protocol (illumination), sensor appearance	Bridges large pixel-statistics gaps by making source imagery target-like prior to training or fine-tuning, when content is preserved [31–34].	Risk of geometry/structure distortion and hallucination; can “fix” the domain gap by altering agronomic signal—needs content preservation constraints and sanity checks.
Test-time adaptation + entropy minimization; calibration/ensembles	None	Protocol (mild sensor/scene)	Fast, label-free plug-in robustness at deployment; calibration/ensembles improve decision reliability and highlight failure modes [6,26,57,58].	TTA can drift without safeguards; calibration/ensembles do not correct systematic bias—use for thresholds/monitoring, not as a substitute for shift-aware evaluation.

Overall, a robust pipeline aligns the robustness taxonomy in Section 3 with the acquisition and sensing choices in Sections 4 and 5, and layers on a graduated adaptation plan: normalize and augment first, add a small and diverse target set for fine-tuning, align features or use teacher–student or related UDA when labels are unavailable, resort to style

translation for large appearance gaps, and stabilize decisions via uncertainty estimation and calibration.

7. Benchmarks and Targeted Gaps: A Practical Agenda for Datasets and Experiments

Robustness under distribution shift is only as credible as the evaluation settings used to measure it. In practice, this depends on which public benchmarks and domain-shift suites are being used, how they are split and analyzed to produce spatially and temporally realistic estimates of performance, and whether newer datasets are designed explicitly for out-of-distribution (OOD) evaluation rather than only for in-distribution accuracy.

Two public benchmarks are particularly relevant in the context of this work. The Global Wheat Head Detection (GWHD) dataset provides diverse, multi-country UAV and ground imagery for wheat head detection and includes explicit cross-site splits, making it suitable to study scene and time shifts in cereal canopies [8,35]. MinneApple offers orchard imagery for fruit detection and counting with substantial variability in illumination, occlusion, and background, which is useful to probe appearance and protocol shifts in tree crops [7]. Related apple datasets for robotic harvesting extend this setting to semantic segmentation and domain adaptation under changing canopy structure and lighting [43]. In addition, several works curate targeted domain-shift benchmarks, such as grape-bunch datasets organized by season, viewpoint, and variety, or cross-site root and soil imaging where texture and moisture vary markedly, as well as multi-site weed and disease datasets that vary in weather and management regimes across fields [14,27,36]. These emphasize combinations of sensor, protocol, scene, and time shifts beyond the random splits used in many earlier studies. Together, these resources support more than in-distribution testing; they allow leave-one-field, leave-one-site, or leave-one-season-out evaluation that reveals transfer gaps, which random tile or patch splits systematically hide [6,7].

To obtain realistic robustness estimates from these datasets, spatially aware partitions or explicit leave-one-site/season-out protocols should be preferred over random splits. Reporting both in-site and OOD scores makes it possible to separate memorization from true transferability [7,8,35]. When metadata permit, results should be reported separately by altitude or GSD and by sun geometry to quantify protocol sensitivity and accompanied by simple calibration checks (such as temperature scaling or binned reliability plots) so that action thresholds remain trustworthy across sites and years [6,26]. For adaptation studies, it is important to fix a reference source model and compare lightweight updates (normalization, small target-set fine-tuning, feature alignment, teacher–student UDA, or style translation) under the same OOD split. This isolates the contribution of the adaptation method from artifacts caused by different split definitions [28,31]. Benchmarks such as GWHD, MinneApple, grape-bunch benchmarks, and multi-site disease datasets already support this design, but many published adaptation papers still rely only on in-domain random splits.

Despite recent progress, several gaps remain and are particularly valuable to fill. First, grain crops beyond wheat are underrepresented in publicly released UAV and proximal RGB datasets with explicit cross-farm and cross-season partitions. Maize, soybean, rice, barley, and sorghum remain especially scarce compared with orchards and wheat, even though several closed or partially released datasets exist for yield, biomass, nitrogen, and suitability estimation [4,9,10,25,38]. Second, there is a lack of cross-sensor benchmarks acquired over the same fields with RGB, multispectral, hyperspectral, and thermal cameras, with measured spectral response functions and radiometric references. Such datasets are needed to study sensor replacement, fusion, and band-compression strategies without confounding changes in the scene, building on calibration-transfer studies that currently

rely on smaller or proprietary collections [23,24]. Third, longitudinal sequences (multi-mission, multi-season) with consistent protocols are still rare, even though they are crucial to quantify temporal drift and to compare small periodic updates against heavier UDA pipelines [5,6,10,35]. Finally, many existing datasets omit practical evaluation add-ons such as standardized panel or PIF frames, exposure and white-balance logs, and wind and temperature notes for thermal missions, even though these details often dominate generalization behavior in the field and are central in recent robustness and calibration-transfer work [9,16].

The reviewed work suggests a compact “benchmark spec” that would make new datasets immediately useful for robustness studies:

1. *Design for OOD*: Include at least three distinct farms or sites and at least one held-out season, and publish recommended leave-one-site/season-out splits alongside conventional random splits [7,8,27].
2. *Protocol cards*: Document UAV flight altitudes and GSD targets, shutter speed, ISO, white balance, panel use, sun conditions, and ground truth procedures; release both raw and standardized imagery together with the exact preprocessing steps, including panel- or PIF-based scaling and any band-compression transforms [5,9,16].
3. *Compact annotations matched to actions*: Provide box or point masks for counting tasks, plot-level labels for management decisions, and inter-annotator agreement samples to bound label noise and support label-efficiency experiments [6,10,12].
4. *Sensor notes and cross-sensor pairs* (optional but impactful): For multispectral, hyperspectral, and thermal data, include band centers and FWHM, emissivity assumptions, and co-registered RGB (and, where feasible, multisensor pairs over the same plots) to enable alignment, calibration-transfer, and band-compression studies [23,24,26].
5. *Reporting template*: Encourage or require spatially aware metrics, explicit OOD scores, calibration plots, and a short description of any adaptation steps so that results remain comparable across studies and across adaptation methods [6,12,35].

Figure 4 summarizes this specification as a reusable benchmark card and links it to a minimal, reproducible experiment suite.

Given current resources, a minimal experiment suite can already make the benchmark agenda more specific and reproducible. (E0) Report an in-distribution (random) split only as a reference. (E1–E2) Run a “baseline honesty” pass that trains a single model and reports both in-site performance and OOD performance under leave-one-site and/or leave-one-season-out evaluation using GWHD, MinneApple, or similar domain-shift splits; optionally add a small ensemble and temperature scaling to assess whether confidence is sufficiently calibrated for decision-making [6–8]. (E4) Add acquisition sensitivity sweeps that systematically vary panel use, exposure locking, and altitude/GSD groupings to quantify protocol sensitivity before introducing more complex architectures [5,9,12]. (E5) Include label-efficiency curves (e.g., 0, 5, 10, and 20 % target labels) that compare no update, small fine-tuning, shallow feature alignment, teacher–student UDA, and style translation under the same OOD split [31]. (E3) Finally, perform cross-sensor stress tests: evaluate RGB-trained models on MS-to-RGB projections (and vice versa) with and without spectral response-aware band transforms, and test whether thermal indices should be normalized or left raw; this isolates sensor mismatch effects and supports concrete comparisons of alignment and calibration strategies [5,23,24,26].

In summary, today’s best public resources—GWHD, MinneApple, and emerging domain-shift suites in grapes, roots, weeds, and diseases—already support realistic, spatially aware evaluation and side-by-side comparisons of adaptation strategies. The most valuable additions are balanced grain-crop datasets (RGB-first, UAV, and proximal) with explicit cross-farm and cross-season splits, longitudinal multi-season collections, and cross-

sensor pairings with full protocol cards. Adopting a minimal benchmark specification of the type outlined above would make robustness claims more comparable and reproducible across sites, seasons, sensors, and management regimes [6–9,14,23,36].

Table 4 distills these requirements into a minimal, reusable specification intended to make robustness claims comparable across crops, sites, sensors, and seasons.

Benchmark card (dataset/spec template)	
Scene card (OOD by design) • Farms/sites ≥ 3; management/cultivar diversity • Seasons/phenology stages annotated (when possible) • Explicit field/site IDs to enable site hold-outs	Sensor card (cross-sensor ready) • Modality (RGB/MS/HS/thermal), camera model • Bands/spectral response (or centers+FWHM) • Radiometry pipeline + compression/preprocessing notes
Protocol card (repeatable acquisition) • Altitude/GSD targets; overlap; view geometry • Time window; exposure/WB policy (locked vs. auto) • Panels/targets used (if any) + weather/illumination notes	Evaluation card (robustness claims) • Provide split files/seeds; prevent near-duplicate leakage • Require at least 1 OOD split (field/farm or season) • Report per-site/per-season + a calibration/uncertainty summary
Minimal robustness experiment suite (E0–E5)	
E0 (ID)	Random split <i>only as reference</i> ; do not claim deployment robustness.
E1 (Scene)	Leave-one-field/farm/site-out split; report per-site breakdown.
E2 (Time)	Leave-one-season/year-out split; quantify drift and update frequency.
E3 (Sensor)	Sensor-out/matched cross-sensor test (paired flights if available).
E4 (Protocol)	Protocol sweep (altitude/GSD tiers, illumination window, panel/WB policy).
E5 (Labels)	Label-efficiency curve (e.g., 1%, 5%, 10%, 30% target labels) for adaptation.

Figure 4. Benchmark agenda made operational. Top: a compact “benchmark card” template specifying the minimum metadata and design choices needed to support shift-aware evaluation in proximal and UAV crop imaging. Bottom: a minimal experiment suite (E0–E5) that connects the four-axis taxonomy to concrete, reproducible, and robustness tests and reporting.

Table 4. Minimal benchmark and reporting specification for shift-aware evaluation in proximal and UAV crop imaging. This checklist turns the “benchmark gaps” discussion into a concrete, reusable template for dataset releases and experimental reports.

Component	What to Report (Minimum)
Data/scene card	Crop, cultivar(s), management regime(s), site description (at least region-level), season(s), phenology stage(s), and known confounders (soil/background, residue, weeds).
Sensor card	Camera model, modality (RGB/MS/HS/thermal), bands/spectral response (or band centers + FWHM), radiometric workflow (DN→reflectance), lens/focal length, and any compression/pipeline steps.
Protocol card	Altitude/GSD, overlap, view geometry, acquisition time window, exposure/WB policy (locked vs. auto), calibration targets/panels (if any), and weather/illumination notes.
Ground truth	Measurement/annotation protocol, sampling design, temporal alignment (image vs. measurement), and uncertainty sources (inter-rater, instrument, timing).

Table 4. *Cont.*

Component	What to Report (Minimum)
Splits	Exact file lists (or seeds + deterministic code) for (i) an in-distribution reference split and (ii) at least one OOD split (field/farm hold-out and/or season hold-out). If claiming sensor robustness, include a sensor-out split on matched plots when feasible.
Metrics	Task metrics plus at least one robustness-oriented summary: per-site/per-season breakdowns, calibration (e.g., ECE/reliability curves), and uncertainty reporting when decisions depend on thresholds.
Baselines	A simple baseline (e.g., RGB or single-sensor) evaluated on the same OOD splits, plus at least one ablation (protocol/radiometry/bandset/adaptation) to support causal claims.
Release (when licensing permits)	Protocol metadata, preprocessing code with fixed seeds, and evaluation scripts; anonymize sensitive location details if needed while preserving split logic.

8. Conclusions

The key conclusions of this work can be summarized as follows:

- Robustness failures in proximal and UAV crop imaging can be interpreted along four shift axes (scene, sensor, protocol, and time), making failure modes diagnosable rather than generic.
- Acquisition and standardization choices (solar windows, timing, exposure/white balance, radiometry, GSD/overlap, and metadata) often dominate apparent gains from model changes.
- Credible robustness claims require spatially and temporally aware evaluation (e.g., field/farm/site and season/year hold-outs); random splits are mainly in-distribution references.
- Adaptation is most effective when staged and symptom-driven, ranging from lightweight normalization/augmentation to fine-tuning and alignment matched to the dominant gap.

Robustness under distribution shift is emerging as a central obstacle to deploying proximal and UAV imaging models in real agricultural settings. Across the 42 core studies reviewed here, a consistent picture emerges in which models do not fail “in general”, but in specific, diagnosable ways along four axes: scene, sensor, protocol, and time. Scene changes alter canopy geometry, background clutter, and management regimes; sensor changes bring different spectral response functions, resolutions, and noise properties; protocol differences reshape view geometry, illumination, and radiometry; and temporal drift across phenological stages, seasons, and years compounds these effects. Treating these axes explicitly turns robustness from a vague objective into a concrete design space for acquisition, evaluation, and adaptation.

This review also complements and extends existing surveys on UAV and proximal phenotyping, crop monitoring, and agricultural machine learning. Prior reviews have mainly organized the literature by trait (for example, disease detection [62], yield prediction [63], or nitrogen status [64]), by sensor or platform (e.g., broad proximal/remote phenomics overviews [65], UAV [66] versus proximal systems [50]), or by model family (such as convolutional networks [67] or transformers [68] for crop imaging) and have typically evaluated methods under in-distribution settings. Complementary perspective papers in plant pathology argue that many reported gains do not translate to deployment because training datasets are often not representative of real field variability (backgrounds, devices, cultivars, growth stages and lighting), which inflates apparent performance and obscures generalization limits [50,51]. Likewise, generic surveys on domain adaptation or transfer learning in remote sensing seldom focus on field-scale crop monitoring or on the

concrete acquisition and evaluation constraints faced in agronomic experiments. Related system-level reviews in precision weeding and directed-energy robotics emphasize that real-world performance depends on an end-to-end perception–decision–actuation stack (not just model accuracy), with practical constraints such as field capacity, automated navigation, energy efficiency, affordability, and safety standards shaping what is deployable. We cite these deployment-oriented lessons as complementary context for why our benchmark agenda stresses protocol disclosure, shift-aware splits, and reproducible reporting for proximal and UAV crop imaging [69]. By contrast, the present review takes robustness under distribution shift as its primary organizing principle and proposes a field-focused taxonomy along four axes (scene, sensor, protocol, and time). It then uses a systematically defined corpus of 42 studies to link this taxonomy to acquisition protocols, evaluation designs, adaptation and updating strategies, and benchmark specifications. The goal is not to catalog every phenotyping pipeline but to provide a coherent framework and a practical decision aid for researchers and practitioners who need models that continue to work as farms, sensors, and seasons change.

The empirical evidence summarized in Sections 3 and 4 shows that a well-designed RGB baseline, acquired and evaluated carefully, already generalizes more broadly than many practitioners expect. When ground sampling distance, view geometry, and illumination are controlled, RGB models for head, fruit, and stress detection can maintain performance across fields and seasons. Multispectral, hyperspectral, and thermal modalities add robustness primarily when the trait of interest is explicitly physiological (nitrogen status, chlorophyll, fuel moisture, water stress) or when lighting and background are unstable in ways that RGB alone cannot compensate. In those settings, red-edge and NIR bands, physics-guided spectral features, and canopy temperature relative to ambient conditions dampen appearance shifts and stabilize trait estimation across sites, sensors, and years. However, additional bands only improve robustness when radiometry, band handling, and cross-sensor alignment are explicit; otherwise, they simply add new avenues for domain shift.

Sections 5 and 6 highlight that robustness is often won or lost before the first model is trained. Acquisition and standardization choices (solar windows, phenological timing, exposure and white-balance control, panel or pseudo-invariant-feature use, GSD and overlap targets, metadata logging) frequently deliver larger gains than changing architectures. Evaluation design is equally fundamental, as random splits at pixel or patch level systematically overestimate transferability by leaking micro-context across folds, whereas spatially aware and leave-one-field/season/site/sensor-out protocols reveal the real generalization gap and provide a realistic baseline for adaptation. On the model side, an incremental adaptation plan, summarized in the roadmap of Figure 2, is more effective than any single technique. Lightweight normalization and augmentation are low-cost first steps that can be deployed even without target labels. When a modest fraction of a new site can be labeled, small, spatially diverse target sets combined with careful fine-tuning of upper layers re-anchor models to new cultivars, backgrounds, or management regimes without retraining from scratch. Shallow feature alignment and calibration-transfer methods stabilize representations across sensors and phenotyping platforms, while teacher–student consistency schemes and related UDA strategies [61] provide label-free improvements when acquisition differences dominate the gap. Style translation can bridge pure appearance shifts, provided content is preserved, and test-time normalization, simple ensembles, and explicit calibration further stabilize predictions and make confidence scores more trustworthy for decision thresholds. Crucially, these techniques are most effective when they are matched to observable symptoms rather than applied as generic recipes.

Section 7 shows that public benchmarks already offer many of the ingredients needed for shift-aware evaluation. GWHD and MinneApple, together with emerging domain-shift benchmarks in grapes, weeds, diseases, and root/soil imaging, support leave-one-site and leave-one-season protocols and explicit cross-sensor or cross-platform tests. At the same time, important gaps remain: grain crops beyond wheat are underrepresented in UAV and proximal datasets with explicit cross-farm and cross-season partitions; cross-sensor suites acquired over the same plots with RGB, multispectral, hyperspectral, and thermal cameras are rare; and longitudinal multi-season datasets with consistent protocols are still the exception rather than the rule. The minimal benchmark specification proposed here is intended as a compact checklist for future dataset releases so that robustness claims become more comparable and reproducible across crops, sites, sensors, and management regimes. The terms “deployability” and “auditability” are framed here as operational goals. The purpose of this specification is to make such claims testable and comparable via shared OOD splits, protocol disclosure, and paired ID/OOD reporting rather than anecdotal demonstrations.

This review has several limitations that should be acknowledged. First, the core corpus is restricted to proximal and UAV imaging studies retrieved by a specific query over a defined set of crops (wheat, rice, maize, soybean, apple, citrus, mango, tomato, grape, and pear) and sensing terms, and only articles written in English were considered. Relevant work in other languages, crops outside this list, or studies indexed under different terminology may therefore have been missed. Second, the focus on title-level robustness terms (e.g., “domain adaptation”, “transfer learning”, “calibration transfer”, “out of distribution”) deliberately concentrates on studies where distribution shift is central but necessarily excludes many otherwise high-quality sensing or modeling articles that do not explicitly highlight these concepts. Third, the included studies are methodologically heterogeneous in tasks, sensors, and evaluation protocols, which precludes any formal meta-analysis of effect sizes. The synthesis here is descriptive and conceptual rather than statistical. Fourth, purely satellite-based works, non-imaging sensor networks, and broader digital-agriculture systems are only touched upon when they inform proximal/UAV pipelines, and thus the conclusions should not be directly extrapolated to those domains. Finally, while the search and coding protocol aimed to be systematic and consistent, it was implemented by a limited number of raters and inevitably reflects judgment calls in borderline cases. These limitations do not undermine the main messages of the review, but they highlight that the taxonomy and recommendations proposed here should be refined and extended as the empirical base on robustness under distribution shift in agricultural imaging continues to grow.

In summary, robustness under distribution shift in proximal and UAV agricultural imaging is less about discovering a single “universal” architecture and more about designing sensing, evaluation, and adaptation pipelines that acknowledge how and why data change across farms and seasons. By making shift axes explicit, standardizing acquisition and evaluation, and matching adaptation tools to observable symptoms, the community can move from ad hoc fixes toward more deployable and auditable pipelines, provided that robustness claims are supported by standardized OOD splits and reporting (as summarized by the benchmark specification and minimal experiment suite).

9. Future Research Directions

Looking ahead, several lines of work appear promising. First, future datasets would benefit from being designed explicitly for out-of-distribution evaluation, with built-in leave-one-out splits, multiple farms or management units, and at least one held-out season, accompanied by protocol data documenting flight timing, radiometry, sensor settings,

and ground-truth procedures (including released split files or deterministic seeds, plus baseline evaluation scripts for direct comparability). Second, cross-sensor and cross-scale benchmarks, where the same plots are imaged with multiple cameras (RGB, MS, HS, thermal) and at multiple scales (proximal, UAV, satellite), would enable systematic studies of sensor replacement, fusion, and band compression without confounding scene changes (ideally with band centers/FWHM or spectral response information and a consistent radiometric workflow). Third, longitudinal and multi-management datasets spanning rotations, management changes, and climate anomalies are needed to compare lightweight periodic updates, heavier UDA pipelines, and purely physics-guided approaches, and to quantify how often models need to be recalibrated or fine-tuned in practice (including explicit update schedules, e.g., per season or per management change, so “how often to update” becomes measurable). Fourth, systematic label-efficiency curves and active-learning strategies under shared OOD splits would help practitioners plan labeling and flight campaigns with clear trade-offs (using standardized target-label budgets such as 0/5/10/20% under the same OOD split). Finally, embedding decision-aware robustness metrics, including calibration, uncertainty, and cost- or risk-based summaries that reflect management actions, would better align technical progress with agronomic and economic value (including simple decision-threshold reporting with recommended operating points when models trigger management actions).

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